



इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND
INFORMATION TECHNOLOGY**

सत्यमेव जयते



Security Operations Center

Management & Best Practices

**Indian Computer Emergency
Response Team (CERT-In)**

What is a SOC?



Centralized unit for monitoring, detecting, and responding to security events

Integrates network defense, endpoint monitoring, cloud visibility, and threat intelligence

Operates 24x7 to maintain situational awareness and response readiness

Acts as a fusion center combining NOC, DFIR, and cyber threat intelligence functions

Key Functions of a SOC

01



Continuous Monitoring

24x7 visibility across endpoints, networks, and cloud.

02



Threat Intelligence Integration

Correlate IOCs and context from global feeds.

03



Threat Detection

To detect and identify potential intrusions or unauthorized activities early.

04



Incident Response

Contain, eradicate, and recover with defined playbooks.

05



Security Analysis & Reporting

To analyze incidents, report findings, and support decision-making.

06



Vulnerability Management

To identify, assess, and fix security weaknesses before exploitation.

07



Automation & Orchestration (SOAR)

To speed response, reduce work, and ensure playbook consistency.

Continuous Monitoring

To provide 24x7 visibility into all digital assets and network activities. SOC analysts use a wide array of monitoring tools to ensure no malicious or abnormal behavior goes unnoticed.

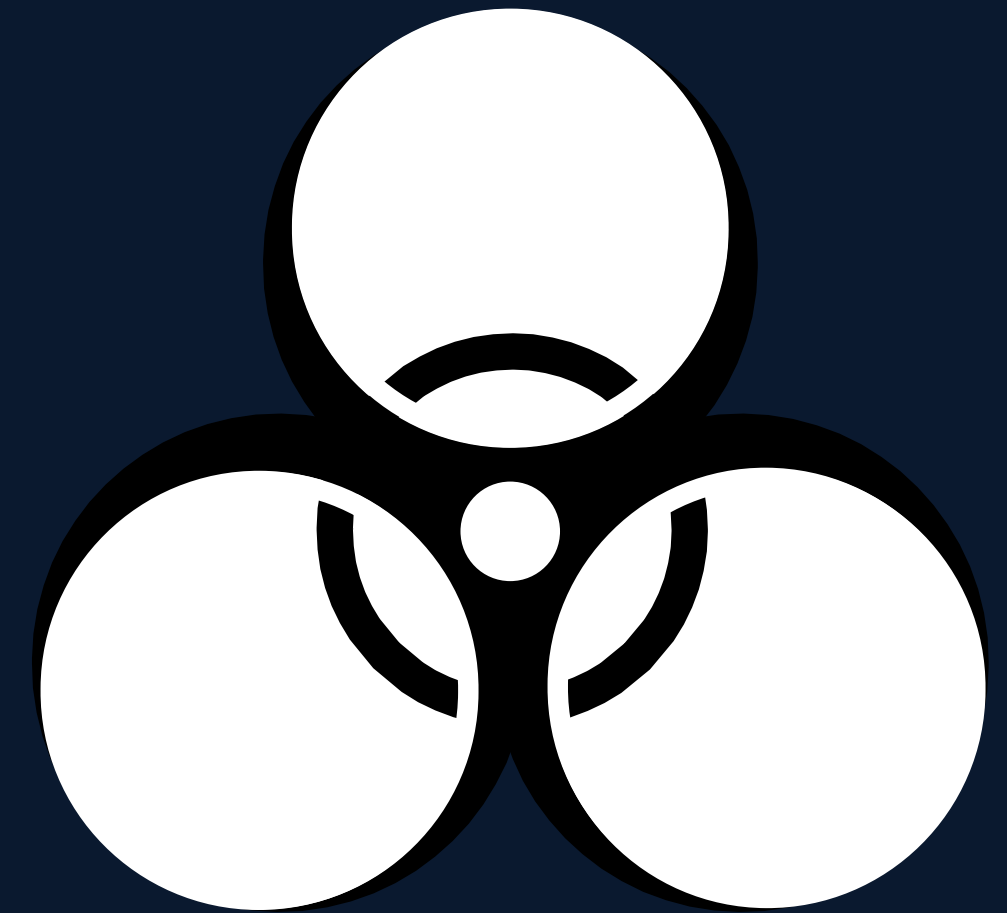
- **Data Collection:** The SOC gathers logs from firewalls, IDS/IPS, EDRs, web servers, network devices, databases, and cloud systems.
- **Centralization:** All logs are sent to a SIEM (Security Information and Event Management) platform that aggregates and normalizes the data.
- **Real-Time Analysis:** Automated correlation rules and machine learning models flag anomalies (e.g., a user logging in from two countries within minutes).
- **Dashboards & Alerts:** SOC analysts receive prioritized alerts to act upon suspicious patterns immediately.



Threat Intelligence Integration

To proactively look for hidden or undetected threats and prepare defenses against evolving adversaries.

- **Threat Intelligence:** Collects and analyzes data from OSINT, commercial, and dark web sources to understand emerging attack vectors.
- **Threat Hunting:** Analysts hypothesize attack scenarios (e.g., “Has anyone used PowerShell to dump credentials in the last 24 hours?”) and actively search telemetry data.
- **Indicators of Compromise (IoCs):** IPs, file hashes, domains, or process names are investigated to uncover hidden intrusions.
- **Hunting Methodologies:**
 - Hypothesis-driven (based on new TTPs)
 - Intel-driven (based on recent campaigns)
 - Anomaly-driven (based on deviations from baselines)



Threat Detection

To detect and identify potential intrusions, compromises, or unauthorized activities before they escalate.

- **Signature-Based Detection:** Matches known threat signatures (e.g., malware hash, command patterns).
- **Behavioral Analysis:** Detects deviations from baseline activity (e.g., data transfer at 3 a.m. from an unusual endpoint).
- **Anomaly Detection:** Uses AI/ML to spot patterns inconsistent with normal operations.
- **Threat Intelligence Integration:** Correlates internal events with external feeds such as AlienVault OTX or MITRE ATT&CK data.
- **Threat Prioritization:** Incidents are scored by severity (critical, high, medium, low) for faster triage.



Incident Response

To contain, mitigate, and eradicate active threats efficiently to minimize damage.

- **Identification:** Analysts confirm whether an alert is a real incident or a false positive.
- **Containment:** Isolate affected hosts, disable compromised accounts, and block malicious IPs.
- **Eradication:** Remove malware, close exploited vulnerabilities, and cleanse affected systems.
- **Recovery:** Restore systems from clean backups and validate operational integrity.
- **Communication:** Coordinate with internal teams (IT, Legal, HR) and external agencies if required.
- **Documentation:** Record the full timeline, decisions, and actions for later review.



Security Analysis & Reporting

To perform in-depth incident analysis, generate detailed reports, and communicate findings to leadership for decision-making and compliance.

- **Forensic Analysis:** Extracts digital evidence (logs, memory dumps, network captures) to reconstruct the attack chain.
- **Root Cause Identification:** Determines how the attacker entered and what damage was done.
- **KPI Tracking:** SOC measures Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).
- **Post-Incident Review (PIR):** Identifies what went right, what went wrong, and what needs improvement.
- **Rule Refinement:** Updates SIEM rules, playbooks, and signatures based on new findings.



Vulnerability Management

To identify, assess, prioritize, and remediate security weaknesses before attackers exploit them.

- **Scanning:** Automated scanners (like Nessus, Qualys, OpenVAS) detect outdated software and misconfigurations.
- **Prioritization:** Vulnerabilities are ranked based on CVSS score, exploit availability, and asset criticality.
- **Patch Management:** Coordinates with IT to deploy patches safely and promptly.
- **Verification:** Confirms remediation success through rescans.
- **Reporting:** Tracks vulnerability trends and overall reduction of attack surfaces.
- **Integration:** Feeds data into SIEM for correlation with threat events.



Automation & Orchestration (SOAR)

To accelerate response, reduce manual workload, and ensure consistent execution of security playbooks.

- **Playbooks:** Predefined workflows automate responses (e.g., blocking IP, isolating endpoint).
- **Integration:** Connects SIEM, EDR, ticketing systems, and firewalls.
- **Automation Use Cases:**
 - Auto-quarantine of infected devices
 - Auto-enrichment of alerts with threat intel
 - Automated notifications to teams.
 - Outcome: Faster response time, fewer errors, better scalability.



Factors which emphasize criticality of SOC

- Identify and protect critical data & assets
- Increase Visibility
- Early threat detection and response
- Reduced downtime and service disruption
- Compliance and regulatory requirements
- Protection of reputation and trust
- Incident analysis and learning
- Increase detection abilities & response capability



SOC Evolution

Gen 1



Gen 2



Gen 3



Gen 4 (Now)



Network-centric
SIEM monitoring

Integrated incident
response and
correlation engines

Intelligence-driven
SOC (TIP, MITRE
ATT&CK mapping)

AI-Augmented
SOC – predictive
analytics, LLM
triage, automation
at scale

Types of SOC

01

02

03

04

05



In-House SOC

Full control and data sovereignty.



Outsourced SOC (MSSP)

Expertise via service providers.



Hybrid SOC

Shared resources and skills development.



Cloud SOC

Native monitoring across multi-cloud environments.



Federated SOC

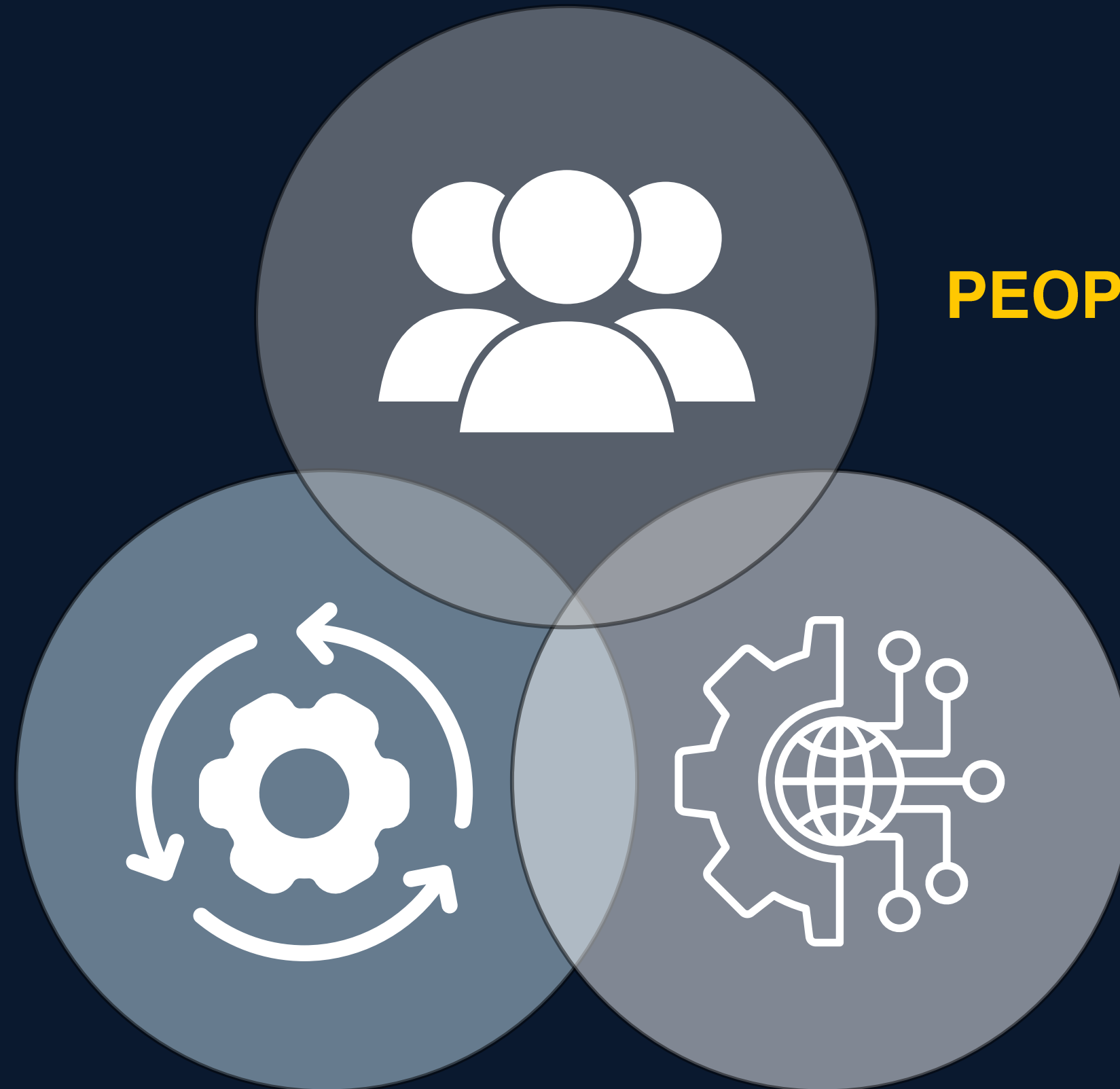
Sectoral or multi-entity coordination model.

SOC Model Comparison

Model	Advantages	Challenges
In-house	Full control, context awareness	Cost & talent retention
Outsourced	Cost efficient, 24x7 coverage	Limited visibility, coordination
Hybrid	Scalability, shared expertise	Integration complexity
Cloud	Elastic resources, modern tooling	Multi-cloud visibility gaps
Federated	Collaboration across entities	Data sharing & governance

Building Blocks of SOC

PROCESS



PEOPLE

TECHNOLOGY

Building Blocks: People

01

SOC Manager

Governance and coordination.

02

Tier 1 Analyst

Alert monitoring & triage.

03

Tier 2 Analyst

Incident analysis & response.

04

Tier 3 Investigator

Forensics & malware analysis.

05

Threat Intel Analyst

Adversary tracking & reporting.

06

Automation Engineer

SOAR scripting & integration.

07

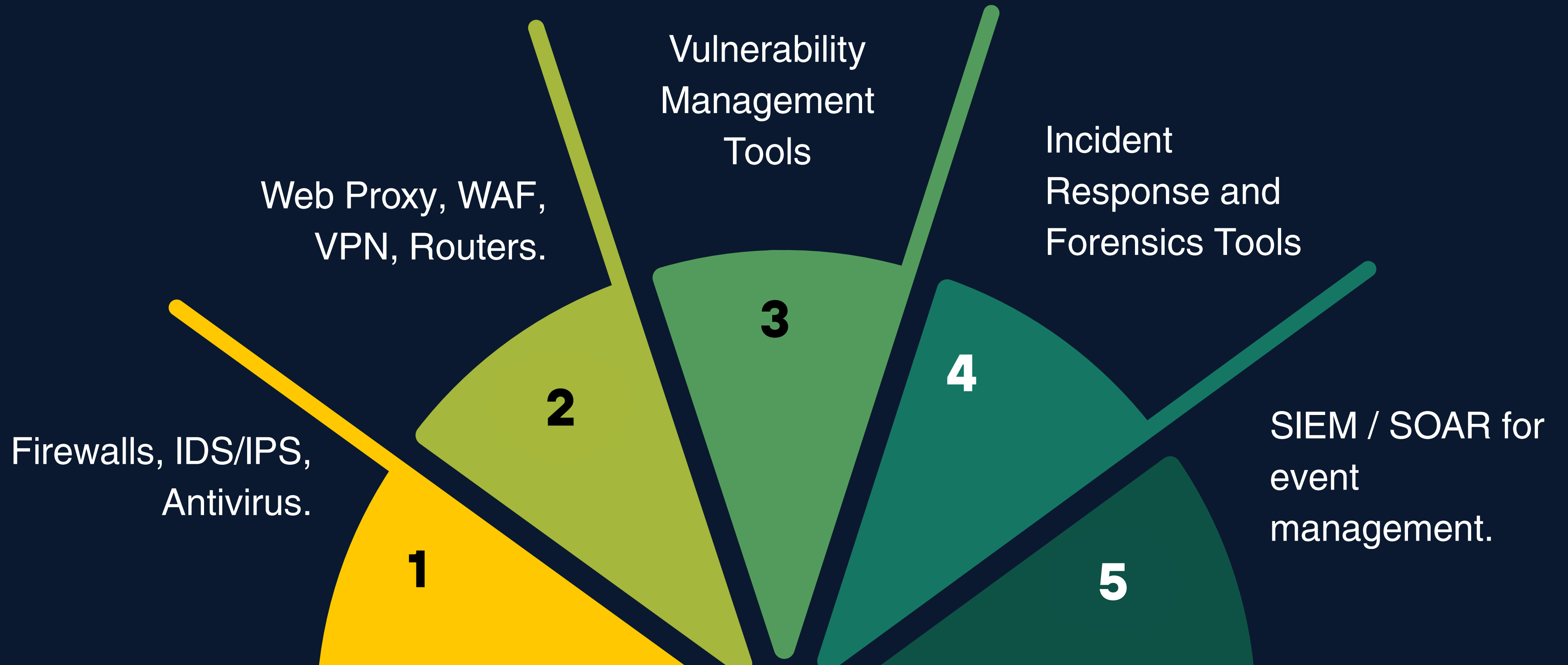
Data Scientist

Behavior analytics & ML models.

SOC Processes



Technology



SIEM & SOAR



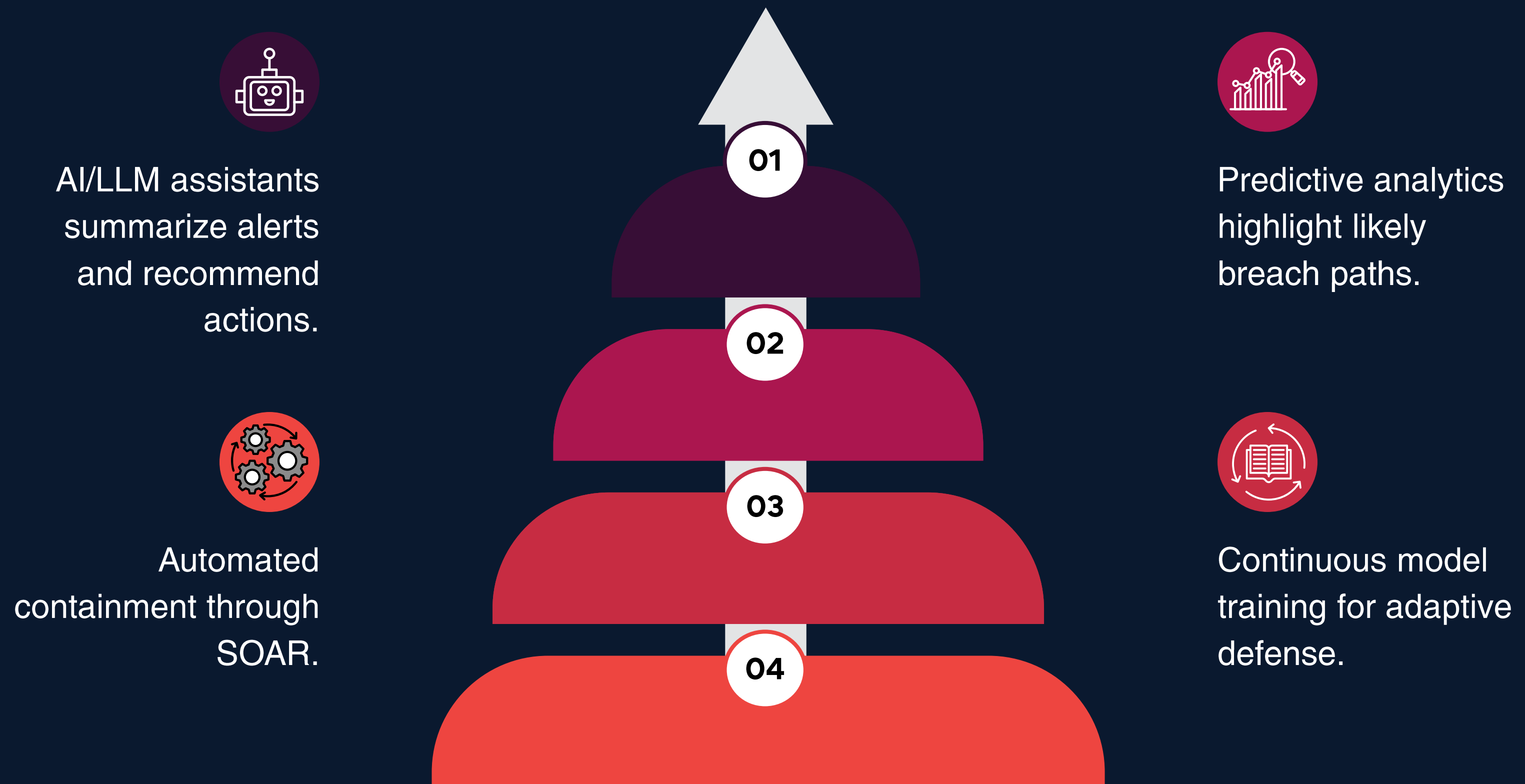
Security Information and Event Management (SIEM)

- Real-time event collection and historical analysis.
- Correlation, alerting, and compliance reporting.
- Log enrichment (Whois, ASN, GeoIP).

Security, Orchestration, Automation, And Response (SOAR)

- Orchestrates data from multiple sources.
- Automates low-level tasks.
- Provides playbooks for incident response.

AI & Automation in SOC



SOC Performance Metrics

Mean Time to Detect (MTTD)

- Time to identify a threat after it occurs.
- Lower = faster detection.

Mean Time to Respond (MTTR)

- Time from detection to containment/remediation.
- Goal: <4 hours for critical incidents.

Mean Time to Investigate

- Time analysts spend from alert triage to root cause confirmation.

Incident Volume

- Total number of incidents handled per month/quarter.
- Aids resource allocation.

True Positive Rate

- % of legitimate alerts among total alerts.

False Positive Rate

- % of alerts that were non-malicious.
- Keep below 10%.

SOC Performance Metrics

Detection Coverage

- % of network, endpoint, and cloud sources monitored.

Automation Rate (SOAR)

- % of incidents automatically handled by scripts or playbooks.
- Boosts speed, reduces workload.

Containment Success Rate

- % of incidents contained before lateral movement or data loss.

Dwell Time

- Duration attackers stay undetected in the system.
- Lower = stronger detection.

Service Downtime due to Security Incidents

- Total time business operations were affected.

Compliance/Audit Readiness Score

- Ability to generate audit logs/reports promptly.
- Reflects operational discipline.

Governance & Compliance

CERT-In Guidelines (2022)

Mandatory incident reporting.

DPDPA 2023

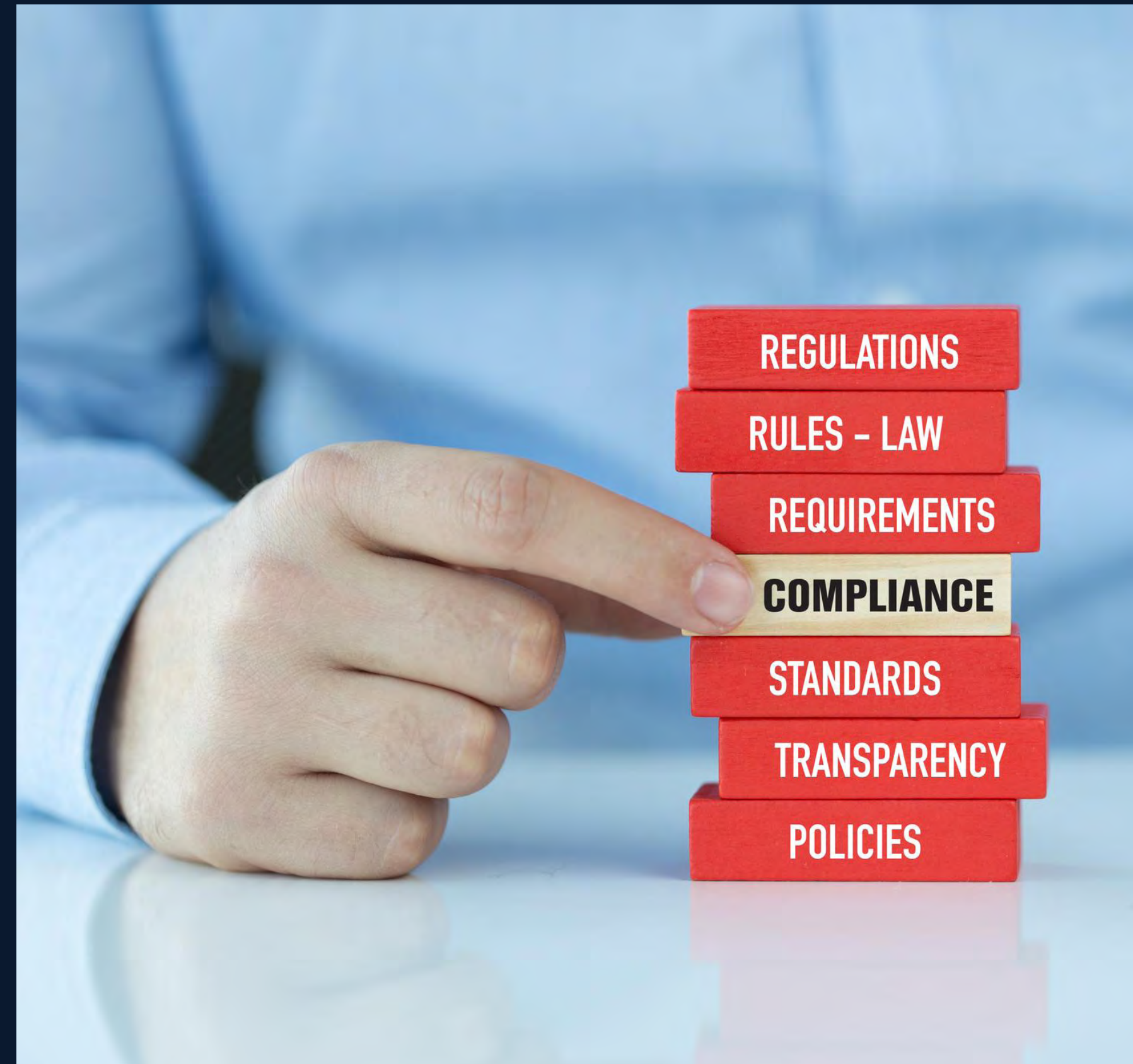
Data privacy and breach notification.

SIM3 / NIST-CSF

SOC maturity benchmarking.

ISO 27035 / NIST 800-61

Incident management frameworks.



SOC Challenges



01



Evolving Threat Landscape.

02



Alert fatigue.

03



Shortage of skilled analysts.

04



AI-driven social engineering and deepfakes.

05



Cross-Team Collaboration.

SOC Maturity Model (SIM3)

Level	Description
0 – Initial	Ad-hoc monitoring and manual processes
1 – Reactive	Basic SIEM alerts and manual response
2 – Proactive	Playbooks and partial automation
3 – Intelligence-Driven	Threat intel integration and hunting
4 – Adaptive	AI automation and continuous improvement

Contact us



24/7 Cyber security Incident Response Help Desk



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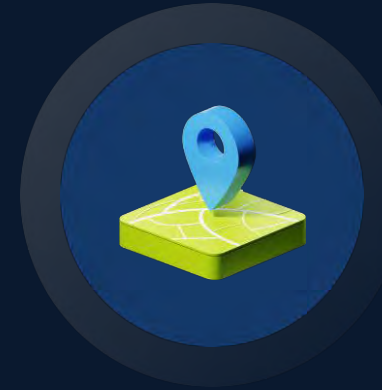
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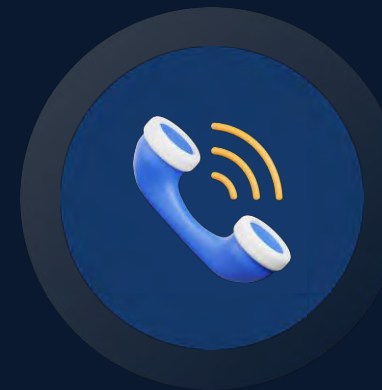
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Thank You

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Criticality of SOC

